Fujisaki compactness for division-algebra adelic units: complete author proof chain

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github:ImperialCollegeLondon/FLT; Apache-2.0

Editorial note. Complete original fixed-version Fujisaki chapter with all core lemmas and delayed main proof. Same native result is an alternative format, not an additional count.

Editorial note. Only print theorem declarations are adapted. One shorthand Haar-character reference is resolved to its original same-work lemma; see the retained source map. Author mathematical text is unchanged.

Allowed standard prerequisites (editorial).

- Additive adelic discreteness/cocompactness and Haar scaling character. same fixed project Adele and HaarCharacter miniprojects. finite-volume quotient and global-unit Haar invariance

Source reference. Original theorem 9: author-original/blueprint/src/chapter/AdeleMiniproject.tex, source lines 1–1. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 10: author-original/blueprint/src/chapter/HaarCharacterProject.tex, source lines 1–1. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 10.13: author-original/blueprint/src/chapter/HaarCharacterProject.tex, source lines 251–251. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 9.47: author-original/blueprint/src/chapter/AdeleMiniproject.tex, source lines 938–938. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 9.51: author-original/blueprint/src/chapter/AdeleMiniproject.tex, source lines 1002–1002. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 10.40: author-original/blueprint/src/chapter/HaarCharacterProject.tex, source lines 904–904. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 9.52: author-original/blueprint/src/chapter/AdeleMiniproject.tex, source lines 1017–1017. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Source reference. Original theorem 10.36: author-original/blueprint/src/chapter/HaarCharacterProject.tex, source lines 788–788. Original named prerequisite navigation; aliased missing shorthand label only if recorded; author mathematical text unchanged.

Miniproject: Fujisaki's Lemma

1 The goal

There is an idelic compactness statement which encapsulates both finiteness of the class group of a number field and Dirichlet's units theorem about the rank of the unit group. In fact there is even a noncommutative version of this statement. In John Voight's book [1] this is Main Theorem 27.6.14(a) and Voight calls it Fujisaki's lemma. I know nothing of the history but I'm happy to adopt this name. In the quaternion algebra miniproject we will use this compactness result to prove finite-dimensionality of a space of quaternionic modular forms.

2 Initial definitions

Let K be a field. A *central simple K-algebra* is a K-algebra B (not necessarily commutative) with centre K such that B has exactly two two-sided ideals, namely 0 and B (or $\bot$ and $\top$, as Lean would call them). We will be concerned only with central simple K-algebras which are finite-dimensional as K-vector spaces, and when K is clear we will just refer to them as central simple algebras. We remark that a 4-dimensional central simple algebra is called a *quaternion algebra*; we will have more to say about these later on.

Matrix algebras $M_n(K)$ are examples of finite-dimensional central simple K-algebras. If $K = \mathbb{C}$ (or more generally if K is algebraically closed) then matrix algebras are the only finite-dimensional examples up to isomorphism. There are other examples over the reals: for example Hamilton's quaternions $\mathbb{H} := \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ with the usual rules $i^2 = j^2 = k^2 = -1$, $ij = -ji = k$ etc, are an example of a central simple $\mathbb{R}$-algebra (and a quaternion algebra), and matrix algebras over $\mathbb{H}$ are other central simple $\mathbb{R}$-algebras. For a general field K one can make an analogue of Hamilton's quaternions $K \oplus Ki \oplus Kj \oplus Kk$ with the same multiplication rules ($i^2 = -1$ and so on) to describe the multiplication, and if the characteristic of K isn't 2 then this is a quaternion algebra (which may or may not be isomorphic to $M_2(K)$ in this generality).

Some central simple algebras B are *division algebras*, meaning that they are division rings, or equivalently that every nonzero $b \in B$ has a two-sided inverse. For example Hamilton's quaternions are a division algebra over $\mathbb{R}$, because $(x + yi + zj + tk)(x - yi - zj - tk) = x^2 + y^2 + z^2 + t^2$, so the inverse of a nonzero $x + yi + zj + tk$ is $(x - yi - zj - tk)/(x^2 + y^2 + z^2 + t^2)$. However 2×2 matrices over a field K, whilst being a central simple algebra over K, are never a division algebra (even if $K = \mathbb{C}$) because a nonzero matrix with determinant zero such as $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has no inverse.

3 Enter the adeles

The adeles of a number field are discussed in far more detail in the adele miniproject 9. We just recall here that if K is a number field then there are two huge commutative topological K-algebras called the *finite adeles* $\mathbb{A}_K^\infty$ and the *adeles* $\mathbb{A}_K$ of K, and that they're both locally compact as topological spaces. We also know from theorem 9.47 that $\mathbb{A}_K \cong K \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}} K$ (both topologically and algebraically), meaning that if R is a K-algebra then $R_{\mathbb{A}} := R \otimes_K \mathbb{A}_K$ is naturally isomorphic to $R \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$. One can furthermore check that if R is a finite K-algebra then the $\mathbb{A}_K$-module topologies and $\mathbb{A}_{\mathbb{Q}}$-module topologies on $R_{\mathbb{A}}$ coincide. Indeed, the topology on $\mathbb{A}_K$ is the $\mathbb{A}_{\mathbb{Q}}$-module topology, as $\mathbb{A}_K = \mathbb{A}_{\mathbb{Q}} \otimes_{\mathbb{Q}} K$ as topological $\mathbb{A}_{\mathbb{Q}}$-algebras, where the right hand side has the $\mathbb{A}_{\mathbb{Q}}$-module topology by definition. So the claim follows from the fact that if A is a topological ring, B is a topological A-algebra finite as an A-module and with the A-module topology, and if M is a topological B-module (and hence a topological A-module), then the A-module and B-module topologies on M coincide (this is `moduleTopology.trans` in the repo, not yet PRed to mathlib).

Let K be a number field and let D/K be a finite-dimensional central simple K-algebra (later on D will be a division algebra (hence the name) but we do not need this yet). Then $D_{\mathbb{A}} := D \otimes_K \mathbb{A}_K$ is an $\mathbb{A}_K$-algebra which is free of finite rank, and if we give $D_{\mathbb{A}}$ the $\mathbb{A}_K$-module topology then it is a topological ring (by results in mathlib). Furthermore $D_{\mathbb{A}}$ is free of finite rank over the locally compact topological ring $\mathbb{A}_K$ and is thus also locally compact. So by the theory of Haar characters (see Chapter 10) there is a canonical character $\delta_{D_{\mathbb{A}}} : D_{\mathbb{A}}^\times \to \mathbb{R}_{>0}$ measuring how left multiplication by an element of $D_{\mathbb{A}}^\times$ changes the additive Haar measure on $D_{\mathbb{A}}$. Let $D_{\mathbb{A}}^{(1)}$ denote the kernel of $\delta_{D_{\mathbb{A}}}$, and give it the subspace topology coming from $D_{\mathbb{A}}^\times$. Corollary 10.40 from the Haar character miniproject shows that $D^\times$ (regarded as a subgroup of $D_{\mathbb{A}}^\times$ via the map $d \mapsto d \otimes 1$) is contained within $D_{\mathbb{A}}^{(1)}$, thus the below theorem typechecks.

Theorem 11.1. *If D is a division algebra then the quotient $D^\times \backslash D_{\mathbb{A}}^{(1)}$ with its quotient topology coming from $D_{\mathbb{A}}^{(1)}$, is compact.*

The rest of this miniproject is devoted to a proof of this theorem.

4 The proof

We prove the theorem via a series of lemmas.

Lemma 11.2. *There's a compact subset E of $D_{\mathbb{A}}$ with the property that for all $x \in D_{\mathbb{A}}^{(1)}$, the obvious map $xE \to D\backslash D_{\mathbb{A}}$ is not injective.*

Proof. We know that if we pick a $\mathbb{Q}$-basis for D of size d then this identifies D with $\mathbb{Q}^d$, $D_{\mathbb{A}}$ with $\mathbb{A}_{\mathbb{Q}}^d$, and $D\backslash D_{\mathbb{A}}$ with $(\mathbb{Q}\backslash\mathbb{A}_{\mathbb{Q}})^d$. Now $\mathbb{Q}$ is discrete in $\mathbb{A}_{\mathbb{Q}}$ by theorem 9.51, and the quotient $\mathbb{Q}\backslash\mathbb{A}_{\mathbb{Q}}$ is compact by theorem 9.52. Hence D is discrete in $D_{\mathbb{A}}$ and the quotient $D\backslash D_{\mathbb{A}}$ is compact.

Fix a Haar measure μ on $D_{\mathbb{A}}$ and push it forward to $D\backslash D_{\mathbb{A}}$; by compactness this quotient has finite and positive measure, say $m \in \mathbb{R}_{>0}$. Choose any compact $E \subseteq D_{\mathbb{A}}$ with measure $> m$ (for example, choose a $\mathbb{Z}$-lattice $L \cong \mathbb{Z}^d$ in $D \cong \mathbb{Q}^d$, define $E_f := \prod_p L_p \in D \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}^\infty$, and define $E_\infty \subseteq D \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^n$ to be a huge

closed ball, large enough to ensure the measure of $E := E_f \times E_\infty$ is bigger than m). Then $\mu(xE) = \mu(E) > m$ so the map can't be injective. □

Definition 11.3. *We let E denote any compact set satisfying the hypothesis of the previous lemma.*

Definition 11.4. *Define $X := E - E := \{e - f : e, f \in E\} \subseteq D_{\mathbb{A}}$.*

Definition 11.5. *Define $Y := X.X := \{xy : x, y \in X\} \subseteq D_{\mathbb{A}}$.*

Lemma 11.6. *X is a compact subset of $D_{\mathbb{A}}$.*

Proof. It's the continuous image of the compact set $E \times E$. □

Lemma 11.7. *Y is a compact subset of $D_{\mathbb{A}}$.*

Proof. It's the continuous image of the compact set $X \times X$. □

Lemma 11.8. *If $\beta \in D_{\mathbb{A}}^{(1)}$ then $\beta X \cap D^\times \neq \emptyset$.*

Proof. Indeed by lemma 11.2, the map $\beta E \to D \backslash D_{\mathbb{A}}$ isn't injective, so there are distinct $\beta e_1, \beta e_2 \in \beta E$ with $e_i \in E$ and $\beta e_1 - \beta e_2 = b \in D$. Now $b \neq 0$ and D is a division algebra, so $b \in D^\times$. And $e_1 - e_2 \in X$ so $b = \beta(e_1 - e_2) \in \beta X$, so we're done. □

Lemma 11.9. *Similarly, if $\beta \in D_{\mathbb{A}}^{(1)}$ then $X\beta^{-1} \cap D^\times \neq \emptyset$.*

Proof. Indeed, $\beta^{-1} \in D_{\mathbb{A}}^{(1)}$, and so left multiplication by β^{-1} doesn't change Haar measure on $D_{\mathbb{A}}$, so neither does right multiplication (by theorem 10.36). So the same argument works: $E\beta^{-1} \to D \backslash D_{\mathbb{A}}$ is not injective so choose $e_1\beta^{-1} \neq e_2\beta^{-1}$ with difference $b \in D$ and then $(e_1 - e_2)\beta^{-1} \in D - 0 = D^\times$. □

Definition 11.10. *Let $T := Y \cap D^\times$.*

Lemma 11.11. *T is finite.*

Proof. It suffices to prove that $Y \cap D$ is finite. But $D \subseteq D_{\mathbb{A}}$ is a discrete additive subgroup, and hence closed. And $Y \subseteq D_{\mathbb{A}}$ is compact. So $D \cap Y$ is compact and discrete, so finite. □

Definition 11.12. *Define $C := (T^{-1}.X) \times X \subset D_{\mathbb{A}} \times D_{\mathbb{A}}$.*

Lemma 11.13. *C is compact.*

Proof. X is compact and T is finite. □

Lemma 11.14. *For every $\beta \in D_{\mathbb{A}}^{(1)}$, there exists $b \in D^\times$ and $\nu \in D_{\mathbb{A}}^{(1)}$ such that $\beta = b\nu$ and $(\nu, \nu^{-1}) \in C$.*

Proof. By lemma 11.8, $\beta X \cap D^\times \neq \emptyset$, and lemma 11.9, $X\beta^{-1} \cap D^\times \neq \emptyset$, so we can write $\beta x_1 = b_1$ and $x_2\beta^{-1} = b_2$ with $b_i \in D^\times$ and $x_i \in X$. Note that $\beta \in D_{\mathbb{A}}^{(1)}$ and $b_i \in D^\times \subseteq D_{\mathbb{A}}^{(1)}$ by corollary 10.40, so $x_i \in D_{\mathbb{A}}^{(1)}$ as well. In particular $x_i \in D_{\mathbb{A}}^\times$ so x_1^{-1} makes sense.

Multiplying the equations defining the x_i and b_i we deduce that $x_2x_1 = b_2b_1 \in Y \cap D^\times = T$ (recall that $Y = X.X$ and $T = Y \cap D^\times$ is finite); call this element t. Then $x_1^{-1} = t^{-1}x_2 \in T^{-1}.X$, and $x_1 \in X$, so if we set $\nu = x_1^{-1} \in D_{\mathbb{A}}^{(1)}$ and $b = b_1 \in D^\times$ then we have $\beta = b\nu$ and $(\nu, \nu^{-1}) \in C := (T^{-1}.X) \times X$. We are done! □

We can now prove Fujisaki's theorem 11.1.

Proof. Indeed, if M is the preimage of C under the inclusion $D_{\mathbb{A}}^{(1)} \to D_{\mathbb{A}} \times D_{\mathbb{A}}$ sending ν to (ν, ν^{-1}), then M is a closed subspace of a compact space so it's compact (note that $\delta_{D_{\mathbb{A}}}$ is continuous, by theorem 10.13, so $D_{\mathbb{A}}^{(1)}$ is a closed subset of $D_{\mathbb{A}}^{\times}$ which is itself a closed subset of $D_{\mathbb{A}} \times D_{\mathbb{A}}$). Lemma 11.14 shows that M surjects onto $D^{\times}\backslash D_{\mathbb{A}}^{(1)}$ which is thus also compact. □

We note here some useful consequences.

Theorem 11.15. *$D^{\times}\backslash(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$ is compact.*

Proof. There's a natural map α from $D^{\times}\backslash D_{\mathbb{A}}^{(1)}$ to $D^{\times}\backslash(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$. We claim that it's surjective. Granted this claim, we are home, because if we put the quotient topology on $D^{\times}\backslash(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$ coming from $(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$ then it's readily verified that α is continuous, and the continuous image of a compact space is compact.

As for surjectivity: say $x \in (D \otimes_K \mathbb{A}_K^{\infty})^{\times}$. We need to extend x to an element $(x, y) \in (D \otimes_K \mathbb{A}_K^{\infty})^{\times} \times (D \otimes_K K_{\infty})^{\times}$ which is in the kernel of $\delta_{D_{\mathbb{A}}}$. Because $\delta_{D_{\mathbb{A}}}(x, 1)$ is some positive real number, it will suffice to show that if r is any positive real number then we can find $y \in (D \otimes_K \mathbb{A}_K^{\infty})^{\times} = (D \otimes_{\mathbb{Q}} \mathbb{R})^{\times}$ with $\delta_{D_{\mathbb{A}}}(1, y) = r$, or equivalently (setting $D_{\mathbb{R}} = D \otimes_{\mathbb{Q}} \mathbb{R}$) that $\delta_{D_{\mathbb{R}}}(y) = r$. But $D \neq 0$ as it is a division algebra,and hence $\mathbb{Q} \subseteq D$, meaning $\mathbb{R} \subseteq D_{\mathbb{R}}$, and if $x \in \mathbb{R}^{\times} \subseteq D_{\mathbb{R}}^{\times}$ then $\delta(x) = |x|^d$ with $d = \dim_{\mathbb{Q}}(D)$, as multiplication by x is just scaling by a factor of x on $D_{\mathbb{R}} \cong \mathbb{R}^d$. In particular we can set $x = y^{1/d}$. □

Remark 11.16. *In this generality the quotient might not be Hausdorff.*

Theorem 11.17. *If U is an open subgroup of $(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$ then the double coset space $D^{\times}\backslash(D \otimes_K \mathbb{A}_K^{\infty})^{\times}/U$ is finite.*

Proof. The double cosets give a disjoint open cover of $(D \otimes_K \mathbb{A}_K^{\infty})$ which descends to a disjoint open cover of the quotient space $D^{\times}\backslash(D \otimes_K \mathbb{A}_K^{\infty})^{\times}$. However this space is compact by theorem 11.15. □

References

[1] John Voight. *Quaternion algebras*, volume 288 of *Graduate Texts in Mathematics*. Springer, 2021. Version v1.0.6u, available at https://jvoight.github.io/quat.html.